Name _____ Period _____ Date _____

Where Did the Old Ocean Floor Go?

Focus question: Why is the seafloor so much younger than the continents?

Warm-up

Which is older, the ocean floor or the continents? Write a guess and a reason.

Pre-check

Glue in: Pre-check (not graded)

Hook data

Glue in: Hook data (D1-D). Then write I notice and I wonder.

Vocabulary

Frayer cards: lithosphere (plate), plate boundary, mid-ocean ridge, subduction, trench

Name _____ Period _____ Date _____

Where Did the Old Ocean Floor Go? (continued)

Focus question: Why is the seafloor so much younger than the continents?

Unit CER page

Glue in: Unit CER page (U-1). Write your first claim.

Plot the pattern

Glue in: World map and site list (D1-A). Write the pattern sentence.

Exit ticket

What pattern do the dots on your map make?

Name _____ Period _____ Date _____

Three Ways Plates Meet

Focus question: What happens where plates meet, and where is ocean crust made and destroyed?

Warm-up

Which pattern did your map dots make? What could explain it?

Warm-up

Vocabulary

lithosphere, plate boundary, mid-ocean ridge, subduction, trench

Plate boundary models

Sketch divergent, convergent, and transform with arrows.

Name _____ Period _____ Date _____

Three Ways Plates Meet (continued)

Focus question: What happens where plates meet, and where is ocean crust made and destroyed?

Boundary table

Glue in: Boundary table (D1-B)

Where is ocean crust made? Where is it destroyed?

Exit ticket

Where is ocean crust made, and where does it sink back down?

Name _____ Period _____ Date _____

Three Ways Plates Meet: Reading Notes

Focus question: How is Earth's surface divided into moving plates?

Warm-up

Skim the headings. What do you predict?

Reading: Lesson 8.1 Earth's Origin (pp. 206-218), first chunk

Biology: The Living Earth. Annotate: underline new terms, star the main idea.

Vocabulary

Word | my definition | example (plate, boundary, ridge, trench)

Name _____ Period _____ Date _____

Three Ways Plates Meet: Reading Notes (continued)

Focus question: How is Earth's surface divided into moving plates?

Main idea in my words

Question I still have

Connect it

Sketch one idea from the reading that helps explain why the ocean floor is young.

Exit ticket

One new idea about plates is ____.

Name _____ Period _____ Date _____

Reading Rock Ages

Focus question: What patterns do the ages of ocean crust and continental crust follow?

Warm-up

A ship pulls up seafloor rock at the Mid-Atlantic Ridge and another near the coast of Brazil. Which is older?

Warm-up

Which rock is older: the ridge sample or the Brazil sample? Say why.

Seafloor ages

Glue in: Seafloor ages (D2-A, D2-B)

Vocabulary

craton (a continent's ancient core)

Name _____ Period _____ Date _____

Reading Rock Ages (continued)

Focus question: What patterns do the ages of ocean crust and continental crust follow?

Graph: age vs. distance

Glue in: Graph grid (D2-C). Plot 7 points, find the rate, predict the age at 3,500 km.

Color North America

Glue in: North America ages (D2-D). Oldest dark brown, youngest light yellow.

Unit CER evidence log

Add two rows: ocean pattern and continent pattern.

Exit ticket

Describe the pattern in ocean crust ages in one sentence.

Name _____ Period _____ Date _____

Reading Rock Ages: Reading Notes

Focus question: How do rocks get dated?

Warm-up

What does it mean for a rock to have an age?

Reading: Lesson 8.1 Earth's Origin (pp. 206-218), continued

Biology: The Living Earth. Annotate: underline new terms, star the main idea.

Vocabulary

Word | my definition | example (radioactive decay, half-life, craton)

Name _____ Period _____ Date _____

Reading Rock Ages: Reading Notes (continued)

Focus question: How do rocks get dated?

Main idea in my words

Question I still have

Connect it

Sketch one idea from the reading that helps explain why the ocean floor is young.

Exit ticket

One new idea about dating rocks is ____.

Name _____ Period _____ Date _____

Clues to Early Earth

Focus question: How do scientists learn about early Earth when the oldest ocean floor is only 200 million years old?

Warm-up

How could anyone know Earth is 4.5 billion years old?

Warm-up

How could anyone know Earth is 4.5 billion years old? Write one idea.

Vocabulary

zircon, radiometric dating, impact crater

45 cm Earth timeline

Glue in: 45 cm Earth timeline (D3-B). Tape your 45 cm strip below.

Name _____ Period _____ Date _____

Clues to Early Earth (continued)

Focus question: How do scientists learn about early Earth when the oldest ocean floor is only 200 million years old?

Evidence list

Glue in: Early Earth evidence log

Unit CER evidence log

Add a row about zircons, rocks, or fossils.

Exit ticket

List two kinds of evidence used to learn about early Earth.

Name _____ Period _____ Date _____

Clues to Early Earth: Reading Notes

Focus question: What does the record of early life tell us about early Earth?

Warm-up

What do you already know about the earliest life?

Reading: Lesson 8.2 Life's Early History (pp. 219-225)

Biology: The Living Earth. Annotate: underline new terms, star the main idea.

Vocabulary

Word | my definition | example (fossil, stromatolite, zircon)

Name _____ Period _____ Date _____

Clues to Early Earth: Reading Notes (continued)

Focus question: What does the record of early life tell us about early Earth?

Main idea in my words

Question I still have

Connect it

Sketch one idea from the reading that helps explain why the ocean floor is young.

Exit ticket

One new idea about early Earth is ____.

Name _____ Period _____ Date _____

Meteorites and Moon Rocks

Focus question: Why do meteorites and moon rocks tell Earth's age better than Earth rocks do?

Warm-up

The Moon and Earth are hit by the same space rocks. Why does Earth show so few craters?

Warm-up

Earth and the Moon are hit by the same space rocks. Why does Earth show so few craters? Predict.

Crater counts

Glue in: Crater counts (D3-A)

Name _____ Period _____ Date _____

Meteorites and Moon Rocks (continued)

Focus question: Why do meteorites and moon rocks tell Earth's age better than Earth rocks do?

Half-life model

Glue in: Half-life model: decay record

How Earth formed

Use meteorites, moon rocks, and Earth's oldest minerals to tell the story.

Unit CER evidence log

Add a row about meteorites, moon rocks, or zircons.

Exit ticket

Why do scientists use meteorites and moon rocks, not Earth rocks, to find the age of the solar system?

Name _____ Period _____ Date _____

Meteorites and Moon Rocks: Reading Notes

Focus question: How old is the solar system?

Warm-up

What is a meteorite?

Reading: Lesson 8.1 Earth's Origin (pp. 206-218), reread for meteorite and moon rock evidence

Biology: The Living Earth. Annotate: underline new terms, star the main idea.

Vocabulary

Word | my definition | example (meteorite, half-life, radiometric dating)

Name _____ Period _____ Date _____

Meteorites and Moon Rocks: Reading Notes (continued)

Focus question: How old is the solar system?

Main idea in my words

Question I still have

Connect it

Sketch one idea from the reading that helps explain why the ocean floor is young.

Exit ticket

One new idea about the age of the solar system is ____.

Name _____ Period _____ Date _____

Why the Ocean Is Young and the Continents Old

Focus question: How is ocean crust made, grown, and recycled, and how do continents grow and survive?

Warm-up

You know the ocean floor is under 200 million years old. Where does the old ocean crust go?

Warm-up

Where does the old ocean crust go?

Guided notes

Glue in: Guided notes (D4-A)

Density lab table

Glue in: Density lab table (D4-B)

Name _____ Period _____ Date _____

Why the Ocean Is Young and the Continents Old (continued)

Focus question: How is ocean crust made, grown, and recycled, and how do continents grow and survive?

SketchNotes: notes

Glue in: SketchNotes checklist (D4-C) above your half page of notes.

Sensemaking: cross-section

Draw: ridge, moving plate, trench, volcanic arc, continent with a younger edge, density labels, carbon arrow.

Unit CER evidence log

Add a row about density or continental growth.

Exit ticket

Why is ocean crust young and why are continents old?

Name _____ Period _____ Date _____

Why the Ocean Is Young and the Continents Old: Reading Notes

Focus question: How are continents formed and why do they last?

Warm-up

What do you know about why continents last longer than ocean floor?

Reading: Lesson 8.3 Earth and Life Evolve Together (pp. 226-235)

Biology: The Living Earth. Annotate: underline new terms, star the main idea.

Vocabulary

Word | my definition | example (density, subduction, continental crust)

Name _____ Period _____ Date _____

Why the Ocean Is Young and the Continents Old: Reading Notes (continued)

Focus question: How are continents formed and why do they last?

Main idea in my words

Question I still have

Connect it

Sketch one idea from the reading that helps explain why the ocean floor is young.

Exit ticket

One new idea about continents is ____.

Name _____ Period _____ Date _____

Evidence for Moving Crust

Focus question: How strong is the evidence that the crust has moved and is still moving?

Warm-up

How could you show that the crust moves?

Warm-up

How could you show the seafloor moved long ago? How could you show it is moving today?

Vocabulary

magnetic reversal

Magnetic stripes

Glue in: Magnetic stripes (D5-A). Tape your paper stripes below it. Label youngest and oldest.

Hawaii and GPS

Glue in: Hawaii and GPS (D5-B). Fill in both rates.

Name _____ Period _____ Date _____

Evidence for Moving Crust (continued)

Focus question: How strong is the evidence that the crust has moved and is still moving?

Pangaea puzzle

Glue in: Pangaea puzzle (D5-C). Tape your best fit above it.

Evaluate the evidence

Glue in: Evaluate the evidence (D5-D), strength 1 to 4.

Exit ticket

Which line of evidence is strongest for past motion? For current motion? Why?

Name _____ Period _____ Date _____

Evidence for Moving Crust: Reading Notes

Focus question: What evidence did scientists use to propose that continents move?

Warm-up

What is your best evidence so far that crust moves?

Reading: Lesson 8.3 Earth and Life Evolve Together (pp. 226-235), second read; Analyzing Data, Extinction Through Time (p. 231)

Biology: The Living Earth. Annotate: underline new terms, star the main idea.

Vocabulary

Word | my definition | example (magnetic reversal, fossil evidence, spreading)

Name _____ Period _____ Date _____

Evidence for Moving Crust: Reading Notes (continued)

Focus question: What evidence did scientists use to propose that continents move?

Main idea in my words

Question I still have

Connect it

Sketch one idea from the reading that helps explain why the ocean floor is young.

Exit ticket

One new idea about the evidence is ____.

Name _____ Period _____ Date _____

Evaluate the Theory, Tell Earth's Story

Focus question: How well does plate tectonics explain the ages of crustal rocks, and what story do ancient materials tell about Earth's formation?

Warm-up

Look back at your Lesson 1 claim. What would you change now?

Sort your evidence

Glue in: Three buckets (D6-E). Sort the rows of your U-1 log.

Part A: argument

Glue in: Part A sentence frames and data bank (D6-A). Write your claim, evidence, reasoning, and limit beside them.

Peer feedback

Glue in: Star and step (D6-D).

Name _____ Period _____ Date _____

Evaluate the Theory, Tell Earth's Story (continued)

Focus question: How well does plate tectonics explain the ages of crustal rocks, and what story do ancient materials tell about Earth's formation?

Part B: Earth's story

Glue in: Evidence cards (D6-B) in order, oldest to newest. Then write the story with the frames (D6-C).

Unit CER evidence log

Mark every row you used today (U-1). Revise your Lesson 1 claim.

Exit ticket

Why is Earth's earliest record missing?

Name _____ Period _____ Date _____

Review Day

Focus question: Which targets do I still need to work on before the CSA?

Warm-up

Which target do you feel least sure about? Why?

Self-rating chart

Glue in: Self-rating chart and work plan (D7-A)

Work plan

List what you will do today.

Name _____ Period _____ Date _____

Review Day (continued)

Focus question: Which targets do I still need to work on before the CSA?

Review stations

Notes from any review station you used.

Vocabulary check

Explain each of the 8 words to a partner without reading.

Exit ticket

The target I will study tonight is ____.

Name _____ Period _____ Date _____

CSA

Focus question: What do you know about the ages of Earth's crust?

Warm-up

Before the CSA, I feel ____ about ____.

What I learned about the ages of Earth's crust

Three things I understand better now.

Where did the old ocean floor go?

Explain it with plates, density, and evidence.

Exit ticket

The one idea I want to remember from this unit is ____.